NISCHAL PRATHIPATI

Mechanical Design Engineer | Product Design, GD&T, FEA and DFM

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PROFESSIONAL SUMMARY

Mechanical design engineer with just under two years of industrial experience at JCB, Caterpillar and Pitti Engineering, spanning off-highway hydraulic and drivetrain systems, powertrain components and EV traction motors. Designs in Siemens NX, SolidWorks and CATIA V5 to BS 8888 and ASME Y14.5, runs structural, thermal and non-linear FEA in ANSYS, and applies DFMEA, PFMEA and DFM to take designs into supplier production. Results include an 18-percentage-point cut in coolant-jacket test failures, a reported 25% fatigue-life extension and an 11% bracket mass saving with a 2.5 factor of safety. MSc Automotive Engineering, University of Leeds. Recent independent projects cover tolerance stack-up, injection-moulding DFM, topology optimisation and reacting-flow CFD.

CORE SKILLS

CAD: Siemens NX, SolidWorks (incl. SolidWorks Plastics), CATIA V5, Creo Parametric, Autodesk Fusion 360 (mesh-to-B-Rep reconstruction)

CAE and CFD: ANSYS Mechanical (structural, thermal, modal, non-linear, topology optimisation), ANSYS Explicit Dynamics, ANSYS Fluent, OpenFOAM

GD&T and standards: ASME Y14.5, BS 8888, 1D worst-case and RSS tolerance stack-up, Cpk-based tolerance allocation

Design and quality: DFMEA, PFMEA, DFM and DFA, injection-moulding DFM, root-cause analysis, Digital Mock-Up and packaging studies, engineering change orders, SPC

Data and PLM: MATLAB, Simulink, Simscape, Python (NumPy), PTC Windchill

Also familiar with: AutoCAD, Abaqus, Creo Simulate, AVL Cruise M, ENOVIA PLM, ISO GPS, Microsoft Office

PROFESSIONAL EXPERIENCE

Junior Mechanical Design Engineer | JCB, United Kingdom — Nov 2025 – Jan 2026

Off-highway machinery: hydraulic and drivetrain subsystems.

- Produced and reviewed 40+ 3D CAD models and manufacturing and assembly drawings for hydraulic components, drivetrain subsystems and global valve assemblies in Siemens NX to BS 8888 and ASME Y14.5 GD&T, contributing to a reported 14% reduction in supplier dimensional errors.
- Managed packaging studies and Digital Mock-Up reviews to resolve spatial and kinematic clashes between hydraulic and drivetrain components; the resulting design countermeasures avoided approximately US$15,000 of pilot-build rework.
- Cut the mass of a cast-iron mounting bracket by 11% by removing material from low-stress regions identified through structural FEA, while maintaining a 2.5 factor of safety under a simulated 5 g shock load.
- Coordinated CAD outputs, design changes and interface information with offshore design teams in India using top-down assembly practice and interface-control checkpoints, reducing design-iteration turnaround time by 15%.
- Performed DFMEA risk assessments to identify potential failure modes early and support the reliability of off-highway machinery.
- Reviewed physical machine-performance data against digital models with Test and Development engineers, informing design modifications reported to improve durability and efficiency.

Graduate Manufacturing Engineer | Caterpillar, United Kingdom — Oct 2024 – Sep 2025

Powertrain components: R&D-to-supplier interface with casting and machining suppliers.

- Investigated high-frequency fatigue failures in powertrain vibration mounts by reviewing failure modes and operating loads, and supported a reinforced isolation design reported to extend component life by 25%.
- Validated a non-linear ANSYS Mechanical model of the mount (hyperelastic elastomer with frictional contact) against hydraulic test-rig displacement data, refining support-stiffness boundary conditions to bring simulated peak deflection within 6% of measurement.
- Coordinated engineering change orders and GD&T tolerance reviews between R&D and external casting and machining suppliers, clarifying drawing requirements and aligning design intent with manufacturable supplier outputs.
- Consolidated test-validation schedules and engineering information into a unified PDM workflow in PTC Windchill, reducing cross-departmental data-retrieval time by approximately 15 hours per week.
- Participated in cross-functional DFMEA workshops for powertrain couplings and contributed to PFMEA and statistical-process-control activities.
- Investigated manufacturing deviations through root-cause reviews and revised CAD models and engineering drawings where required, aligning the documented design definition with manufacturing needs.
- Processed drivetrain and hydraulic engineering data with MATLAB and Python (NumPy) workflows.

Component Engineer | Pitti Engineering Limited, Hyderabad, India Oct 2023 – Jun 2024

EV traction motors and powertrain components: stators, rotors, housings and shafts.

- Reduced the hydrostatic coolant-jacket test failure rate from 22% to 4% (an 18-percentage-point cut, 11 to 2 failures per 50 units) on a liquid-cooled EV traction-motor stator housing by redesigning the O-ring gland geometry (18–22% crush, Ra ≤ 1.6 µm) and tightening counterbore positional GD&T in SolidWorks.
- Raised calculated gravimetric power density by 9% (kW/kg) by cutting subassembly mass by about 8.3% at unchanged rated torque and power, simplifying the end-shield and winding-coil enclosure with DFA, and confirmed end-winding hot spots stayed below the Class H limit (180 °C) with steady-state and transient thermal FEA.
- Redesigned EV motor stator and rotor lamination stacks in SolidWorks and CATIA V5 with FEA support, modifying stack geometry to cut stamping material waste by a reported 15%.
- Tightened journal-bearing clearances from 0.05 mm to 0.03 mm after assessing rotating-assembly vibration in continuous endurance testing, increasing dynamic stiffness and reducing operational NVH by 9 dB.
- Applied PFMEA to precision machining of heavily loaded dynamic shafts, holding runout within 15 µm and reducing post-production rejection rates by 14%.
- Diagnosed a concentricity mismatch during shop-floor validation and delivered immediate tooling countermeasures, preventing approximately US$8,500 of prototype scrap.
- Interpreted 100+ engineering drawings using GD&T and worked with cross-functional teams to resolve drawing ambiguities, reduce build issues and support on-time component delivery.
- Performed structural and transient thermal FEA on high-speed rotating components to validate operating limits and support design sign-off, and applied DFMEA to proposed design changes.

Project Engineer Intern | Bharat Heavy Electricals Limited (BHEL), Visakhapatnam, India Apr 2019 – Jul 2019

Production analysis and quality audits for welded assemblies.

- Analysed production data to identify machine-downtime causes and process inefficiencies, recommending changes associated with a reported 5% throughput improvement.
- Documented non-conformances and supported root-cause investigations into welded-assembly failures during maintenance and quality audits, and prepared inspection-audit reports.

INDEPENDENT ENGINEERING PROJECTS

Self-directed projects | Hyderabad, India Feb 2026 – Present

Self-directed CAD, FEA, DFM and CFD projects built on original geometry after relocating to Hyderabad, alongside a search for Product Design Engineering roles.

1D Tolerance Stack-Up and GD&T Optimisation *(Creo Parametric, Excel)*

- Identified a calculated 0.38 mm axial binding risk in a 7-part hydraulic-drivetrain interface modelled in Creo Parametric, using a closed-loop 1D worst-case and 3σ RSS tolerance stack-up per ASME Y14.5.
- Revised the datum reference frame and moved non-critical shoulders to statistical allocation, securing a calculated 0.15 mm minimum clearance at 85 °C while relaxing those tolerances by 30% (assuming Cpk ≥ 1.33) to cut projected CNC machining cost.

Injection-Moulded Structural Housing (PA66-GF30) *(SolidWorks, SolidWorks Plastics)*

- Designed a PA66-GF30 structural housing for high-vibration service with a uniform 2.5 mm wall, 1.5°–2.0° draft and a 0.6× rib-to-wall ratio.
- Resolved 3 undercut and thin-wall flow restrictions with SolidWorks Plastics DFM diagnostics, converting a multi-slide tool concept into a straight-pull 2-plate mould for an estimated ~20% saving in hard-tooling cost.

Topology Optimisation and Parametric Reconstruction *(ANSYS Mechanical, Fusion 360)*

- Cut the mass of a structural steel heavy-equipment mounting bracket by 29% (1.45 kg per unit) through stress-constrained topology optimisation, holding a 2.15 factor of safety (162 MPa peak stress) under simulated 3-axis hydraulic shock loads.
- Rebuilt the optimised facet mesh as manufacturable parametric B-Rep CAD with R6 mm machinable fillets, raising the simulated first natural frequency by 16% to 142 Hz.

Reacting-Flow CFD Validation: Hydrogen-Air Detonation *(OpenFOAM)*

- Validated a transient OpenFOAM reacting-flow model of stoichiometric hydrogen-air detonation (1D/2D shock tube, 9-species Ó Conaire mechanism, 0.1 mm resolution) against NASA CEA and Caltech Chapman-Jouguet data, matching CJ velocity within 2.85% and CJ pressure within 4.43%.

EDUCATION

MSc Automotive Engineering | University of Leeds, United Kingdom 2022 – 2023

Upper Second-Class Honours (2:1), 63% overall. Dissertation: Enhancing EV Performance through Suspension System Optimisation.

BTech Mechanical Engineering | Vellore Institute of Technology (VIT), Vellore, India 2018 – 2022

First Class, CGPA 7.93 / 10.

RESEARCH AND ACADEMIC PROJECTS

MSc Dissertation: Enhancing EV Performance through Suspension System Optimisation

University of Leeds | 2022 – 2023

- Improved simulated vehicle stability by 15% through iterative optimisation of suspension component geometry and active-suspension control parameters on an EV platform.
- Built multi-body dynamics models of the vehicle and suspension in MATLAB, Simulink and Simscape, and designed the suspension components in CAD.
- Quantified handling and stability through virtual vehicle-dynamics tests, including constant-radius cornering, stability-index analysis and static and dynamic load cases.
- Verified the structural integrity of suspension components with FEA in ANSYS, and applied DFMEA and DFA during virtual prototyping to screen failure modes and simplify assembly.
- Communicated results through concise data analyses and technical reports.

Autonomous Transportation Pod (Connected and Autonomous Vehicle), MSc group project

University of Leeds | 2022 – 2023

- Designed a 3000 mm × 2000 mm ladder-frame chassis in CAD for a 10-occupant (1000 kg) autonomous pod with wheelchair and luggage space, and contributed to the exterior body design.
- Simulated frontal impacts at 40 km/h and 70 km/h in ANSYS Explicit Dynamics to evaluate the chassis structural response under crash loading.
- Integrated the chassis with the powertrain, battery, steering and sensor subsystems into a single vehicle concept.
- Contributed to selecting a Bosch eAxle and a 39.4 kWh LiFePO4 battery pack, a wireless-charging concept, a steer-by-wire architecture fusing camera, LIDAR and radar inputs, and a weighted materials decision matrix (aluminium, stainless steel, POM, carbon fibre) with component cost analysis.

F1 Aerodynamics Comparative Analysis, BTech project

VIT

- Showed a simulated 12% downforce increase and 6% drag reduction by proposing a 1.8° rake angle, revised wing angles of attack and optimised sidepods, diffuser and endplates on a conceptual 2022-regulation F1 car.
- Modelled the car to 2022 FIA technical regulations in CATIA and SolidWorks and compared its aerodynamic behaviour in turbulent air with 2021-generation cars.
- Ran CFD in ANSYS Fluent and OpenFOAM to resolve flow structures, downforce and drag, refining the geometry iteratively from the simulation results.

Spur-Gear Pair Design for an Electric Vehicle

University of Leeds | Mar 2023

- Designed spur-gear pairs for an EV transmission, defining gear geometry, ratio selection and dimensional tolerances, and checked tooth bending and contact stresses under simulated operating loads with ANSYS Mechanical FEA.

Waste Plastic to 3D-Printer Filament Machine, BTech project

VIT

- Designed an integrated machine concept that converts waste PP, PS, PET and HDPE into 3D-printer filament, modelling the assembly (hopper, reducer, extruder) in SolidWorks.
- Defined a seven-stage process (jaw and cone crushing, shredding, induction melting with a turning screw, piston-pump transfer, water-coil cooling, extrusion and sticky-roller finishing) and sized a 10 HP induction motor with a 1:30 reducer for a 150 kg batch.
- Built the business case with a cost analysis (materials, electricity, registration, labour), projected revenue and a SWOT analysis.

Regenerative Braking System (conceptual), BTech project

VIT

- Developed a regenerative-braking concept in which a 450 W DC motor-generator recovers kinetic energy during braking to recharge the battery, specifying a 200 mm brake drum, brake lining and linkage mechanism.

Bhavishya Auto Labs: Vehicle Testing Centre Business Plan, BTech project

VIT

- Planned a chain of government-regulated vehicle inspection and certification centres (brake, speedometer, suspension and emissions testing), projecting a 4-lane centre cost of INR 730 lakh, annual revenue of INR 4.03 crore, break-even in 1.8 years and a 10.47% two-year ROI.

STUDENT ENGINEERING TEAMS

Mechanical Design Engineer | Team Eco Titans, VIT (Shell Eco-marathon, IC Engine Prototype) Aug 2020 – May 2021

- Designed lightweight powertrain-mounting brackets in Fusion 360 and ANSYS, achieving a simulated 24% bracket mass reduction through multi-load-case FEA while holding a minimum 2.0 factor of safety under peak cornering and braking loads.
- Contributed to SolidWorks subsystem packaging, ANSYS structural validation and safety-compliance documentation for the 2021 Shell Eco-marathon Virtual Technical Inspection, with first-pass approval reported for the chassis, steering and powertrain-mounting subsystems.
- Supported chassis hardpoint integration and steering clearance checks around the powertrain mounts.

Design Engineer | Team Celerity Solar EV, VIT Oct 2019 – Feb 2022

- Led the aerodynamic bodywork design in CATIA, SolidWorks and ANSYS CFD and FEA, delivering a reported 15% reduction in aerodynamic drag.
- Optimised the chassis with CAD and ANSYS FEA for a reported 10.5–12% weight reduction while maintaining structural integrity.
- Managed bill-of-materials creation and led testing and data analysis, contributing to a 4th-place ranking in the Indian region and a top-50 overall placing at Shell Eco-marathon (2020–21 virtual season, UrbanConcept).
- Applied DFM, floor planning and process planning with the manufacturing team, shortening assembly times and achieving a reported 20% improvement over the team's previous production process.
- Modelled the powertrain in MATLAB and Simulink to maximise driving range, and processed telemetry in MATLAB to correlate multi-body models against accelerometer data for suspension tuning and structural hardpoints.
- Created component designs and detailed manufacturing drawings that balanced cost, performance and safety.

CERTIFICATIONS

- Gold Certification, Electric Vehicles and Renewable Energy, NPTEL (IIT Madras)
- Battery Pack Design for Electric Vehicles, Udemy (instructor: Mohan Reddy, PhD, IISc)
- Two-Wheeler Design and Manufacturing, TVS Limited
- Product Lifecycle Management, Great Learning Academy
- CAD/CAM, Vellore Institute of Technology

LANGUAGES AND ADDITIONAL INFORMATION

Languages: English (C1–C2; IELTS Academic, 2022), Telugu (native), Hindi (C1, professional working), German (A1, elementary)

Mobility: Indian driving licence (car and motorcycle) and International Driving Permit. Open to relocation within India and internationally, and to travel for test-track, supplier and tooling-site visits. Available to start immediately.